Lab 3: Component Modelling & Architectural Pattern Selection — Written Justification and Component Diagram

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Architecture Selection: "I chose Layered Architecture for the Hospital Bed & ICU Allocation Dashboard System."

Architectural Choice:

The Layered Architecture was selected, organizing the system into three layers: a Presentation Layer (Dashboard UI Component) that hospital staff and admins interact with; a Business Layer containing the Bed Allocation Manager, ICU Recommendation Engine, Authentication & Access Control, and Alert & Notification components, which implement the core allocation, recommendation, and alerting rules; and a Data Layer (Bed & Patient Database Component) that persists bed status, patient, and allocation data.

Two Reasons for This Choice:

- **Separation of allocation logic from UI and storage:** FR-001 to FR-005 require real-time bed viewing, filtering, allocation, ICU recommendation, and status/alert management. Isolating these rules inside dedicated Business Layer components keeps them independent of how the dashboard is presented or how data is stored, so allocation or ICU-matching policy can change without touching the UI or database schema.
- **Fit for a single, centrally operated deployment:** the dashboard serves one hospital's wards and ICU rather than many independently scaled services, so a Layered Architecture gives the needed separation of concerns while avoiding the network latency, data-consistency, and operational overhead that a Microservices split would add for this scope.

Security Advantage:

Routing every request for patient, admission, and bed-allocation data through a dedicated Authentication & Access Control component and a single Data Layer component gives the system one consistent, auditable point at which to enforce TLS encryption in transit, encryption at rest, and role-based access restricted to authorized hospital staff, satisfying NFR-002 without duplicating security logic across multiple independent services.

Performance Benefit:

Confining all persistence to the Bed & Patient Database Component lets the team apply targeted optimizations — such as indexing bed and ICU availability records and caching frequently requested availability queries — directly at the data-access layer. This keeps bed and ICU availability lookups fast and predictable, supporting the NFR-001 requirement of loading current availability within 2 seconds under 500 concurrent users, without the added network hops between independently deployed services.

Component Diagram:

